

The surface area of a prism

Two bases and one rectangle — the net says the formula better than the formula does.

LESSON 23

1 × 40 MIN

GEOMETRY 11, §11

IGCSE E13.X

LESSON CLOCK

8'

12'

10'

10'

5'

The net	8 min
The formula	12 min
Oblique prisms	10 min
Working backwards	10 min
Homework	5 min

LEARNING OBJECTIVES

- Derive the lateral surface area of a right prism from its net.
- Compute the total surface area of a prism.
- Use the perpendicular section for an oblique prism.
- Solve for a missing dimension from a given surface area.

TEXTBOOK

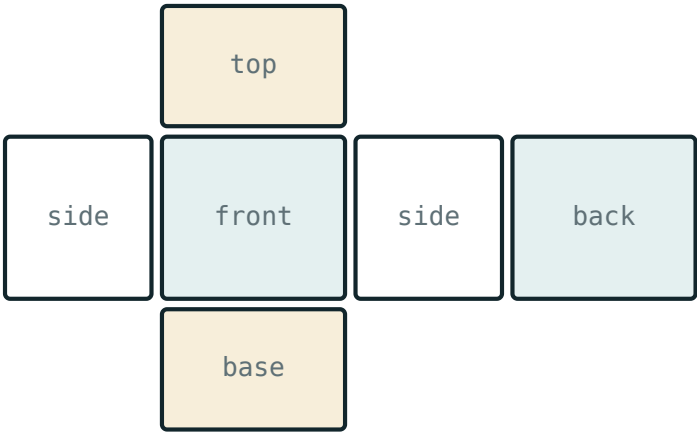
National	pp. 113–118
Cambridge	Core & Extended, pp. 297–300
Workbook	Exercise 11.1

01

Explanation

The net

Cut a right prism along one lateral edge and roll the sides flat. The lateral surface is a single rectangle: its height is the prism's height, and its width is the base perimeter.



surface area = the area of the whole net

The lateral surface unrolls into one rectangle, P wide and h tall.

$$S_{lat} = P \cdot h$$
$$S_{total} = 2B + P h$$

where B is the base area and P the base perimeter.

Worked shapes

PRISM	B	P	S_{TOTAL}
cuboid $a \times b \times h$	ab	$2(a + b)$	$2ab + 2(a+b)h$
cube, edge a	a^2	$4a$	$6a^2$
regular triangular, edge a , height h	$\frac{\sqrt{3}}{4} a^2$	$3a$	$\frac{\sqrt{3}}{2} a^2 + 3ah$
regular hexagonal, edge a , height h	$\frac{3\sqrt{3}}{2} a^2$	$6a$	$3\sqrt{3} a^2 + 6ah$

TWO BASES, NOT ONE

A closed prism has a lid. $S_{total} = 2B + Ph$: the 2 is the mark most often lost. An open container — a trough, a box without a lid — needs the formula adjusted, and the question will say so.

Oblique prisms

For an oblique prism the lateral faces are parallelograms of different shapes, and the net is not one rectangle. The formula generalises by using the **perpendicular section** — the cut at right angles to the lateral edges:

$$S_{lat} = P_{\perp} \cdot \ell$$
 where ℓ is the lateral edge

For a right prism the perpendicular section is the base and $\ell = h$, so the general formula reduces to the one above.

02 Worked examples

EXAMPLE 1 A regular hexagonal prism has base edge 4 cm and height 10 cm. Find its total surface area.

$$① \quad B = \frac{3\sqrt{3}}{2} \times 16 = 24\sqrt{3} \approx 41.57$$

$$② \quad P = 24$$

$$③ \quad S_{lat} = 240$$

$$④ \quad S = 2(41.57) + 240 \approx 323.1$$

ANSWER

$$48\sqrt{3} + 240 \approx 323 \text{ cm}^2$$

EXAMPLE 2 A cuboid has surface area 136 cm² and base 4 × 5. Find its height.

$$① \quad 2(20) + 2(9)h = 136$$

$$② \quad 18h = 96$$

$$③ \quad h = \frac{16}{3} \approx 5.33$$

ANSWER

$$\frac{16}{3} \approx 5.33 \text{ cm}$$

EXAMPLE An oblique prism has lateral edge 12 and perpendicular section a triangle with sides 3, 4, 5. Find the lateral area.

① $P_{\perp} = 12$

② $S_{lat} = 12 \times 12$

③ $= 144$

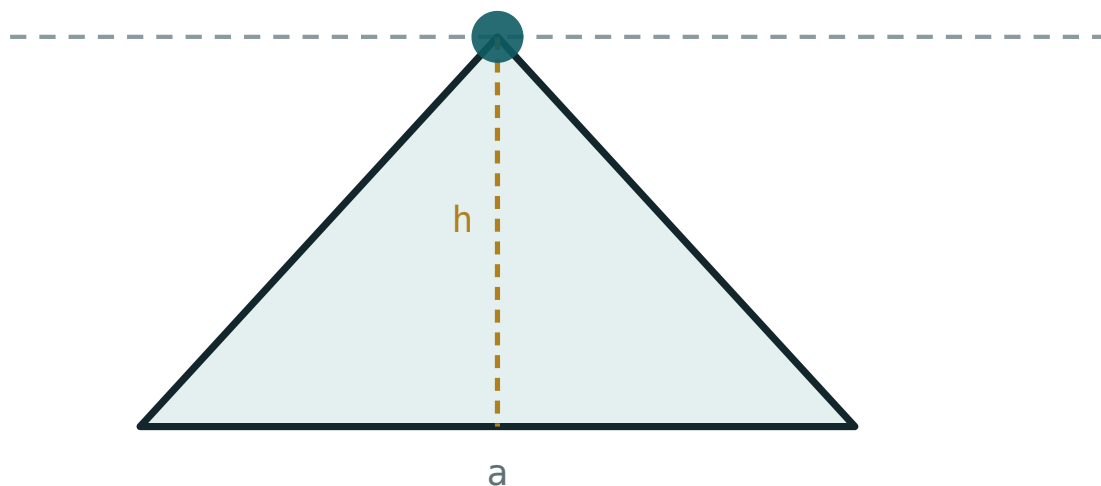
ANSWER

144 square units

03 Interactive model

Unroll a cardboard tube or box and measure the rectangle directly.

Area from a net

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Lateral surface area	Yon sirt yuzasi	Площадь боковой поверхности
Total surface area	To'la sirt yuzasi	Площадь полной поверхности
Net	Yoyma	Развёртка
Base perimeter	Asos perimetri	Периметр основания
Base area	Asos yuzasi	Площадь основания
Perpendicular section	Perpendikulyar kesim	Перпендикулярное сечение
Lateral edge	Yon qirra	Боковое ребро
Apothem of a polygon	Ko'pburchak apofemasi	Апофема многоугольника

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

S_{lat} of a right prism is:

S_{total} of a right prism is:

A cube of edge 4 has surface area:

For an oblique prism S_{lat} uses:

the base perimeter

the perpendicular section perimeter

the height

the diagonal

An open box needs:

$2B + Ph$

$B + Ph$

Ph

$2B$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Surface area of a cube of edge 3

ANSWER 54

2. Surface area of a cube of edge 7

ANSWER 294

3. Cuboid $2 \times 3 \times 4$

ANSWER 52

4. Cuboid $5 \times 5 \times 10$

ANSWER 250

5. S_{lat} of a prism with $P = 20$, $h = 6$

ANSWER 120

6. S_{total} with $B = 15$, $P = 16$, $h = 5$

ANSWER 110

7. A cube has surface area 150. Find its edge

ANSWER 5

Medium · the standard question

7 PROBLEMS

1. Regular triangular prism, edge 6, height 10: total area

ANSWER $18\sqrt{3} + 180 \approx 211$

2. Regular hexagonal prism, edge 4, height 10: total area

ANSWER $48\sqrt{3} + 240 \approx 323$

3. Cuboid with base 4×5 and area 136: the height

ANSWER $\frac{16}{3}$

4. Right prism, base a 3-4-5 triangle, height 9: total area

ANSWER $12 + 108 = 120$

5. Oblique prism, edge 12, perpendicular perimeter 12: lateral area

ANSWER 144

6. Open cuboid $3 \times 4 \times 5$ (no lid): area

ANSWER 82

7. A cube of surface area 384: its space diagonal

ANSWER $8\sqrt{3}$

Hard · stretch and reason

7 PROBLEMS

1. A cuboid has square base and total area 294 with height equal to the base edge. Find the edge

ANSWER 7

2. Regular hexagonal prism of base edge a and height a : total area

ANSWER $3\sqrt{3}a^2 + 6a^2$

3. A prism has a rhombus base with diagonals 6 and 8, height 10. Total area

ANSWER $48 + 200 = 248$

4. A cuboid of volume 240 and square base 4: total area

ANSWER 272

5. Two similar prisms with areas 75 and 300: the length ratio

ANSWER 1 : 2

6. A prism's lateral area equals its total area minus what?

ANSWER twice the base area

7. Prove $S_{lat} = P_{\perp} \ell$ for an oblique prism

ANSWER Each parallelogram has base a side of the perpendicular section

07 Homework

Homework — 4 tasks

Say whether the solid is closed or open before choosing the formula.

1. A regular hexagonal prism has base edge 6 cm and height 15 cm. Find its total surface area.
2. A cuboid with base 6×8 has total surface area 376 cm^2 . Find its height.
3. A right prism has a 5-12-13 triangular base and height 20. Find its total surface area.
4. An open trough is a cuboid $2 \times 3 \times 8$ with no lid. Find the area of material needed.